

Identification of p-phenylenediamine in hair dye samples

Name: Sana Tilak

Department: Department of Chemistry, Azim Premji University

Course: UGRD-31072-2 CHM-Chemicals in the Community

Professor: Sathish CG

Date: November 26, 2025

Abstract

There has been quite a bit of research done recently regarding the carcinogenic effects of hair dye on the general public. India is currently a growing market for hair dyes and many individuals use these products regularly across age groups for a variety of reasons. It is important for consumers to be aware of what are the effects caused by the items they are consuming.

Hair dyes contain p-phenylenediamine which when reacted with hydrogen peroxide in a basic environment can produce Bandrowski's base, a mutagenic base capable of causing cancer. In this experiment, four hair dyes have been analysed to detect which among them have p-phenylenediamine. This experiment also compared the concentrations of p-phenylenediamine present among all four hair dyes.

Our hypothesis was that all the hair dyes will contain p-phenylenediamine. The results for comparing the concentrations of p-phenylenediamine among the four samples were determined through spectrophotometric analysis in the visible light spectrum.

The experiment showed that all four hair dye samples—Streax, Indica, Garnier, and Godrej—contained p-phenylenediamine. The Garnier hair dye had the highest concentration of p-phenylenediamine while Godrej had the least. It can be inferred that Godrej is the safest hair dye to use in this study, while Garnier is the least.

Although there is no concrete evidence to prove that humans face a high risk of contracting cancer when exposed to hair dyes that contain p-phenylenediamine, there has been research which shows Bandrowski's base having mutagenic effects on bacterial and eukaryotic cells. There are safer alternatives in the market which omit chemicals like p-phenylenediamine in their hair dye recipe, so it is recommended to use those.

Introduction

This experiment aims to determine the variance in concentration of carcinogenic material in hair dyes – specifically four different hair dyes commonly found in the Indian market: Streax, Indica, Godrej, and Indica. The compound suspected to cause cancer through hair dyes is known as Bandrowski's base. Bandrowski's base is a structure formed due to the oxidation of p-phenylenediamine, through catalysis by agents such as hydrogen peroxide. There will be more on the formation mechanism in the coming sections.

The purpose of this experiment was to determine firstly there was presence of p-phenylenediamine, and secondly which hair dye possessed greater concentration when compared among each other. The reason p-phenylenediamine was chosen as a candidate to be tested for rather than hydrogen peroxide was because hydrogen peroxide was an ingredient clearly stated in the ingredient label of the hair dyes. P-phenylenediamine was a more elusive ingredient, with some packaging not even including the entirety of the ingredient label.

Another important fact to note is that p-phenylenediamine is the limiting reagent in the reaction where it reacts with hydrogen peroxide to form Bandrowski's base. Since p-phenylenediamine is what controls the amount of Bandrowski's base and the hair dye packets all had the same amount of hair dye in total, it was therefore more appropriate to determine the presence of this reagent and compare its concentrations within the four hair dyes used in the experiment. (All the hair dyes had clearly stated the presence of hydrogen peroxide so there was no confusion whether it existed as an ingredient)

I hypothesize that all the hair dyes will contain p-phenylenediamine since it is a very common ingredient in hair dyes. However, I will only be able to compare the concentrations once I perform visible light spectroscopy on the hair dye samples.

The equipment used in this experiment were measuring cylinders, beakers, and pipettes (5mL and 10mL). The only specialized apparatus utilised was the portable PASCO spectrometer which allowed us to compare absorbance values to determine which hair dye had the highest concentration of p-phenylenediamine when compared with the others.

The Indian hair dye market is currently valued at 0.66 billion USD and is projected to grow till 1.72 billion USD by 2030. It is important to conduct research on topics which will benefit the greater population and prevent them from taking decisions that will be detrimental to their wellbeing. This paper aims to shed light on some of the toxins present in hair dyes and which prominent hair dyes in the market are plausible to contain more than others. (Researchandmarkets 2025)

Experimental Procedure

Some key points to keep in mind before the explanation of the procedure:

The reaction mechanisms behind the formation of Bandrowski's base and how it is mutagenic.

Step 1: Oxidation of p-phenylenediamine

P-phenylenediamine is oxidized by hydrogen peroxide to form p-benzoquinone diimine. P-benzoquinone is a reactive intermediate. Reactive intermediates are important in the formation of products. P-benzoquinone is made up of two conjugated C=O groups that are directly attached to an aromatic ring. The carbonyl groups present withdraw electron density and this causes the ring to be strongly electrophilic. This strong electrophilic nature of p-benzoquinone ensures that it reacts soon.

Step 2: First coupling reaction

The protonated form of p-benzoquinone diimine reacts with an unoxidized p-phenylenediamine molecule through a coupling mechanism. This creates an N-phenyl-p-phenylenediamine dimer (also known as 2,4',5-triaminodiphenylamine).

Step 3: Second coupling

The dimer is attacked by another molecule of p-benzoquinone diimine. This results in a trimeric intermediate which is Bandrowski's base in its reduced form.

Step 4: Final oxidation

The trimer is oxidised to yield Bandrowski's base. During this process, it is also to be noted that a molecule of *p*-phenylenediamine is regenerated. It is surmised that this molecule behaves as a catalyst for the final conversion from the diimine intermediates to Bandrowski's base.

Bandrowski's base: A trimer with a quinone core, with amino groups attached at the 2- and 5-positions, with 4-aminophenyl substituents attached to both the imine nitrogens.

Chemical name: N',N'-bis(4-aminophenyl)-2,5-diamino-1,4-quinonediimine

Mutagenic nature:

Reactivity:

The structure of Bandrowski's base includes reactive quinone imine and aniline moieties (a moiety is a part of a molecule) which can form adducts (product formed due to the direct combination of two or more molecules, where all the molecules are combined into a single product) with nucleic acids and proteins within cells. This leads to mutations.

Experimentally found evidence:

Bandrowski's base has been shown to produce mutagenic activity in tests conducted with bacterial and eukaryotic cells, so the suspicion of it being carcinogenic is not unfounded.

Mechanisms that can facilitate the mutagenic activity of Bandrowski's base:

1. Quinone amines have the ability to participate in redox cycling and generate reactive oxygen species. Reactive oxygen species can lead to oxidative damage in DNA.

2. Aminophenyl groups can facilitate covalent binding to DNA bases (usually guanine), and this can cause base mispairing and hence cause mutations. (Ames et al. 1975)

Method:

Note: The words “about” and “approximately” are used frequently in the procedure. This is not to indicate that the measurements were taken accurately. The method was performed as accurately as possible with the materials at the lab’s disposal, but there is always a chance for minute errors.

1. Preparation of Sample

About 1.00 gram of hair dye liquid was weighed for each hair dye brand respectively and transferred into a 50 mL beaker. To this beaker, approximately 20.0 mL of distilled water was added and the mixture was stirred using a magnetic stir bar at a low temperature. A defined mass and volume was taken for each sample to ensure that the results were consistent and comparable. Water helps in the dissolution of ionic and polar components of the hair dye – which aids in bringing the *p*-phenylenediamine into the solution. The heating at very low temperature (around 40 degrees Celsius) helps increase solubility and dissolution rate of the substances without degrading thermally sensitive compounds such as *p*-phenylenediamine.

2. Alkaline extraction of p-phenylenediamine

Approximately 10.0 mL of sodium hydroxide solution was added to the dye solution and stirred using a magnetic stirring bar for five minutes. After this, about 10.0 mL of ethanol was added to the solution and was also stirred for five minutes or until the solution looked like it was

completely mixed. The sodium hydroxide converts p-phenylenediamine into its deprotonated form (more basic). This essentially improves the extraction of p-phenylenediamine from the dye matrix and stabilizes it. The addition of ethanol also helps similarly – with creating a mixed aqueous-organic medium that improves the solubility of p-phenylenediamine and helps separate it from the dye matrix.

3. Filtration to remove insoluble solids

The mixture was filtered through filter paper into a clean beaker in case there were any undissolved solids that needed to be removed. The filtered mixture was collected in a clean beaker to avoid any contamination. Insoluble particulates could interfere with the spectrophotometric measurements and ensure that the observed color change is only due to dissolved p-phenylenediamine.

This procedure (Steps 1-4) were to be repeated for each of the four hair dyes: Streax, Indica, Godrej, and Garnier. It was also ensured that all the hair dyes were of the same colour (in this case black) to ensure uniformity.

4. Colorimetric detection using Folin-Ciocalteu reagent

Approximately 5.0 mL of the filtrate was transferred into a clean test tube. 1-2 drops of Folin-Ciocalteu reagent were added to the test tube and the tube was mixed gently. The test tube was left to stand for ten minutes at room temperature. The colour change was noted. All the samples were blue, but Godrej was the lightest and the others were of similar colour. Hence, it is confirmed that all samples have the presence of p-phenylenediamine.

The Folin-Ciocalteu reagent works by oxidising the electron-rich p-phenylenediamine molecules which leads to the formation of a blue chromophore (part of a molecule that is responsible for its color by absorbing certain wavelengths of visible light while reflecting others) whose intensity indicates the amount of p-phenylenediamine present.

5. Quantitative spectrophotometric analysis

We did not have p-phenylenediamine in the Chemistry lab, so we could not determine the exact concentration of p-phenylenediamine in the hair dye extract from the measured absorbance. Due to this, we compared the absorbances of all four hair dyes at the same wavelength. The greater the absorbance determines the greater concentration. The explanations and results will be explained in greater detail in the following section.

Results and discussion

For good spectrophotometric analysis, we diluted each sample with approximately 5.0 mL of water. This is so that the concentration of the sample can be reduced to a level that falls between the spectrophotometer's optimal linear range → essentially it ensures that the absorbance values being recorded are accurate.

The reason why more absorbance indicates higher concentration can easily be explained by the Beer-Lambert Law. ($A = \epsilon \cdot c \cdot l$) In this equation A is the absorbance that is measured at a specific wavelength, ϵ is the molar absorptivity (a constant for a given substance at a certain wavelength), c is the concentration of the solution, and l is the path length of the cuvette (it is 1 cm for the cuvettes in the university's Chemistry lab). When concentration is increased, there are more molecules to absorb the incident light, and this leads to a higher absorbance value. Hence when measured at a constant wavelength, with the same parameters used, absorbance and concentration is proportional.

Results:

<i>Hair Dye Sample</i>	<i>Wavelength (nm)</i>	<i>Absorbance</i>
Streax	387.1	1.538
Indica	387.1	1.507
Godrej	387.1	0.438
Garnier	387.1	1.890

Conclusion

As observed in the table above the hierarchy of concentration of p-phenylenediamine in the hair dye samples from greatest to lowest is as follows → Garnier, Streax, Indica, Godrej. Streax and Indica have similar absorbance values, while Garnier and Godrej have a noticeable difference when compared to the other two.

Our hypothesis of there being p-phenylenediamine in all the samples was proven to be correct. According to our results, Godrej is the safest hair dye to use and Garnier is the most potent. The reason why p-phenylenediamine concentration matters more than hydrogen peroxide is because it is the limiting reagent when reacted with hydrogen peroxide to form the mutagenic Bandrowski's base. Although there isn't definitive human evidence to prove that hair dyes containing p-phenylenediamine are carcinogenic, it is safer to steer clear of hair dyes which contain it. There are quite a few safer alternatives in the Indian hair dye market which don't contain p-phenylenediamine like *Vegetal Bio Colour*, *Indus Valley*, and *Deyga Organics*.

References

1. Saranya, L., Gurupadayya, B., Kinnera K, & Thejaswini, J. (2014). Spectrophotometric determination of P-Phenylenediamine in hair dyes. In *JSS University, JSS College of Pharmacy, Department of Pharmaceutical Analysis* (Vols. 11–3, pp. 295–306) [Journal-article].
<https://www.turkjps.org/pdf/68f48462-babf-415b-b878-1e392d1c83af/articles/12335/295-306.pdf>
2. Archana, G., Pawnesh, P., Tanya, S., & Taniya, D. (2023). Determination of paraphenylenediamine in hair dyes formulation using visible spectroscopy. *International Journal of Pharmaceutical Quality Assurance*, 14(04), 917–920.
<https://doi.org/10.25258/ijpqa.14.4.17>
3. BenchChem Technical Support Team. (2025). *The Formation of Bandrowski's Base: A Deep Dive into the Mechanism of p-Phenylenediamine Oxidation*.
https://www.benchchem.com/pdf/The_Formation_of_Bandrowski_s_Base_A_Deep_Dive_into_the_Mechanism_of_p_Phenylenediamine_Oxidation.pdf
4. Bijkersma-Pot, L. (2014). P-phenylenediamine: Insights in penetration, sensitization and elicitation. In *University of Groningen*.
https://pure.rug.nl/ws/portalfiles/portal/14265709/Chapter_2_.pdf
5. PubChem. (n.d.). *Bandrowski's base*. PubChem.
https://pubchem.ncbi.nlm.nih.gov/compound/Bandrowski_s-base
6. Ames, B. N., Kammen, H. O., & Yamasaki, E. (1975). Hair dyes are mutagenic: identification of a variety of mutagenic ingredients. *Proceedings of the National Academy of Sciences*, 72(6), 2423–2427. <https://doi.org/10.1073/pnas.72.6.2423>

7. Ltd, R. a. M. (n.d.). *India Hair Color Market Size, Competitors & Forecast to 2030*.

Research and Markets Ltd 2025.

<https://www.researchandmarkets.com/report/india-hair-dye-market?srsltid=AfmBOooxF>

w2eA7sRX2qqWGp3R3pLRK11jN_nmqMRr7ezQMmelVHDUxRL